

8. (a) Keeping warm / 88 W

1A 1

$$(b) \quad R_1 = \frac{V^2}{P} = \frac{220^2}{88} \\ = 550 \, \Omega$$

1M

1A 2

$$(c) \quad \text{Total current } I_0 = \frac{P_0}{V} = \frac{550}{220} = 2.5 \, \text{A} \\ \text{Current in } R_1, I_1 = \frac{220}{550} = 0.4 \, \text{A} \\ \text{Current in } R_2, I_2 = 2.5 - 0.4 = 2.1 \, \text{A}$$

Or	Power to R_2 , $550 \, \text{W} - 88 \, \text{W} = 462 \, \text{W}$
	Current in R_2 , $I_2 = \frac{P_2}{V} = \frac{462}{220}$
	$= 2.1 \, \text{A}$

1M

1M

1A 3

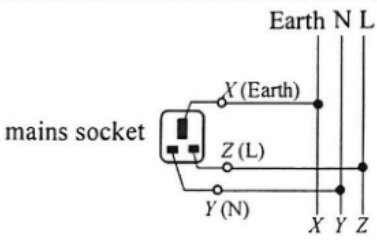
$$(d) \quad \text{Peak current} = \sqrt{2} (2.5 \, \text{A}) \\ = 3.54 \, \text{A}$$

1M

1A 2

8. (a) $P = \frac{V^2}{R}$ $500 = \frac{220^2}{R}$ $R = 96.8 \, \Omega$	1A 1	Accept 97 Ω
(b) As the total resistance of the circuit for mode X is doubled, $\text{total power dissipation} = \frac{V^2}{2R}$ $= \frac{220^2}{2 \times 96.8} = 250 \, \text{W}$ <i>Alternative method:</i> As the voltage across each heating element (1 and 2) is halved, power dissipated in each of the heating element $P_1 \text{ or } P_2 = \frac{500}{4} = 125 \, \text{W} \text{ (since } P \propto V^2 \text{)}$ Or $P_1 \text{ or } P_2 = \frac{V^2}{R} = \frac{110^2}{96.8} = 125 \, \text{W}$ Or $i = \frac{V}{R_1 + R_2} = \frac{220}{2 \times 96.8} = 1.14 \, \text{A}$ $P_1 \text{ or } P_2 = i^2 R = 1.14^2 \times 96.8 = 125 \, \text{W}$ Total power dissipation = $2 \times 125 \, \text{W} = 250 \, \text{W}$	1M 1A 1M 1M 1M 1A	Accept 249 W for 97 Ω
(c) In mode Z, the equivalent resistance of the heating elements is the least as they are connected in parallel, hence, under the same voltage, the total power dissipation is the largest since $P = \frac{V^2}{R}$.	2 1A 1A 2	
(d) (i) For mode Z, the total power dissipated = $500 + 500 = 1000 \, \text{W}$ $I_z = \frac{P}{V} = \frac{1000}{220} = 4.55 \, \text{A}$ <i>Alternative method:</i> $R_{\text{eq}} = \frac{96.8}{2} \, \Omega = 48.4 \, \Omega$ $I = \frac{220 \, \text{V}}{48.4 \, \Omega} = 4.55 \, \text{A}$ Most suitable value of fuse = 5 A	1M+1M 1M+1M 1A 3	
(ii) Although the heater still works in either connection, it is dangerous for switch S to be fitted in wire B (neutral) as the heater / cable would still be live even when the switch was turned off.	1A 1A 1+1 2	
(iii) Wire C (Earth). Current would be conducted from the case through this wire to the Earth.	1A 1A 2	

8. (a) (i) To terminals X	1A	
	1	
(ii) $P = IV$ $800 \text{ W} = I(220 \text{ V})$ $I = 3.636364 \text{ A}$ $\approx 3.64 \text{ A}$	1M 1A	
	2	
(iii) $800 = \frac{V^2}{R} + \frac{V^2}{4R} = \frac{5V^2}{4R}$ $P_{\text{keep warm}} = \frac{V^2}{4R}$ $= 800\left(\frac{1}{5}\right) = 160 \text{ W}$	1M 1M 1A	
<u>Alternative (I)</u> $800 = \frac{V^2}{R} + \frac{V^2}{4R} = \frac{5V^2}{4R}$ $R = 75.625 \Omega$ $P_{\text{keep warm}} = \frac{V^2}{4R}$ $= \frac{220^2}{4(75.625)}$ $= 160 \text{ W}$	1M 1M 1A	
<u>Alternative (II)</u> Power $\propto \frac{1}{\text{resistance}}$ $\frac{P_{\text{keep warm}}}{P_{\text{heating}}} = \frac{R_{\text{eq}}}{4R}$ $P_{\text{keep warm}} = P_{\text{heating}} \left(\left(\frac{1}{R} + \frac{1}{4R} \right)^{-1} / 4R \right)$ $= 160 \text{ W}$	1M 1M 1A	
	3	
(b) (i) Electrical energy (consumed)	1A	
	1	
(ii) (1) only the fuse blows	1A	
(2) only the RCCB cuts off the supply	1A	
	2	

8. (a) 	1A	
(b) (i) - If one of the lighting sets / circuits fails, the other (in parallel) can still operate, i.e. both work independently. - Both can work at the rated power. - Any reasonable answer	1A Any ONE	1
(ii) $P = IV$ $(300 + 450) = I(220)$ $I = 3.409091 \text{ A} \approx 3.41 \text{ A}$ Thus 5 A fuse should be used.	$I = \frac{P_1}{V} + \frac{P_2}{V} = \frac{300}{220} + \frac{450}{220}$	1M 1A 1A 3
(c) Electrical energy used per day $= 0.500 \text{ kW} \times 8 \text{ h} + 2 \text{ kW} \times 0.5 \text{ h} + 3 \text{ kW} \times 2 \text{ h}$ $= 11 \text{ kW h}$ Cost = \$0.9 / kW h $\times$ 11 kW h $= \$9.9$	1M 1M 1A	3

8. (a) Because L_1 and L_2 (connected in series) are shorted by BCDE

1A

Accept p.d. across each low-beam headlight (L_1 or L_2) is 0.

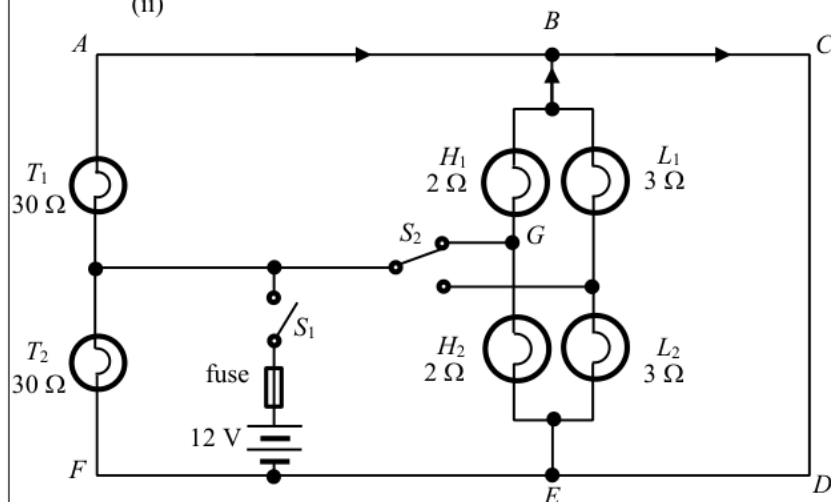
1

(b) (i) p.d. across $T_2 = 12$ V

1A

1

(ii)



largest current in branch BC

1A

1A

Any two currents correctly marked.
All currents correct.

1A

3

(c) Power supplied $P = 2 \times \left(\frac{12^2}{30} + \frac{12^2}{2} \right)$
 $= 153.6 \text{ W} = 154 \text{ W}$ (3 sig. fig.)

1M

1A

1M for same voltage across each bulb (T_1 , T_2 , H_1 and H_2)
e.c.f. from (b)(i)

$$\frac{V^2}{R_{eq}} = \frac{12^2}{R_{eq}} = 153.6 \quad \left(\text{or } \frac{1}{R_{eq}} = \frac{1}{30} + \frac{1}{30} + \frac{1}{2} + \frac{1}{2} \right)$$

1M

1A

1M for all bulbs (T_1 , T_2 , H_1 and H_2) in parallel

$$R_{eq} = 0.9375 \Omega \approx 0.938 \Omega \approx 1 \Omega \quad (3 \text{ sig. fig.})$$

4

(d) max. current $I_{\max} = \frac{V}{R_{eq}} = \frac{12}{0.9375} = 12.8 \text{ A}$

1M

e.c.f. from (c)

$$\left(\text{or } I_{\max} = \frac{P}{V} = \frac{153.6}{12} = 12.8 \text{ A} \right)$$

1A

1A for 12.8 A and correct conclusion

Since the rating is slightly higher than the max. current (when high-beam headlights and taillights are switched on), it is suitable.

2